

Progress Report

DataGear: Data Protection Education

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Over the past months, our team has strategically implemented a thorough and iterative human-centered design (HCD) process to develop DataGear, our digital security awareness tool. Our user-centric approach guided us from initial conceptualization of DataGear all the way through prototyping and user testing. Being receptive to user feedback ensures that our solution meets the needs of our target users, primarily younger generations that often use technology significantly in their day-to-day lives—but due to time constraints our user testing encompasses anyone interested in protecting their data online.

We began our development process by creating personas like “Jared”, a middle schooler who watches videos and TV shows on the internet, plays online games, and uses a Chromebook for school, yet has little knowledge of digital risks. These personas helped us visualize the types of users we were designing for and keep our design decisions relevant to real-life experiences. Next, we created a survey that would reveal how people currently approach digital security and what barriers they face. We received responses from 29 participants across multiple generations, including Gen Z, Gen X, Millennials, and Boomers. The quantitative portion of our survey allowed us to assign cybersecurity awareness scores based on five key behaviors. We used Power BI to build visualizations like a decomposition tree and small multiple bar charts, which helped us break down these scores by generation and behavior. This enabled us to quickly identify trends, such as younger users engaging in risky behaviors like using public Wi-Fi for sensitive tasks without antivirus or VPN protection. These visuals were key to spotting disconnects between self-reported proficiency and actual cybersecurity practices. We found that many users, especially Gen Z, rated themselves as technically advanced but still lacked core knowledge like data protection rights or an understanding of what tools such as VPNs and incognito mode actually achieve. This revealed a gap between confidence and real cybersecurity awareness.

The qualitative portion of our survey added context to our quantitative findings. We conducted open coding on participant responses to questions like "What is the biggest challenge you face in protecting your personal data online?" There were commonly repeated feelings such as confusion, feeling overwhelmed, or not knowing where to start. These were then analyzed through open and axial coding methods where we identified three overarching themes: information overload, gaps in knowledge and accessibility, and awareness and critical evaluation. These themes were visualized using an affinity diagram to uncover patterns in digital behavior and its relationship with informational barriers.

We used these themes to shape our design priorities and create general user personas. These included users who are overwhelmed by digital privacy settings, users who are curious but misinformed, and users who frequently go online with limited knowledge of digital risks. With those personas in mind, we began working on our wireframes and prioritized simplicity, clear navigation, and interactivity to address user preferences for content that was direct and concise. Our designs include quizzes, videos, and readings to accommodate numerous learning styles and preferences.

After a round of user testing using our wireframes we received invaluable feedback reinforcing what we did well and opportunities to improve. Testers appreciated the clear and easy-to-navigate layout, accessibility features, and the diverse learning elements. At the same time, we learned that identified areas for improvement: better guidance for users who may not know where to start, and better explaining concepts and their relevance to protecting themselves online. Throughout this design process, we reflected on our user feedback and findings to ensure our project follows HCD practices from the beginning to the end. Our process closely followed the Double Diamond UX framework. In the discovery phase, we collected qualitative and quantitative data to explore the problem space without assumptions. In the define phase, we coded user responses and built affinity diagrams to highlight themes like information overload and gaps in knowledge. During the development phase, we used this insight to create personas, wireframes, and learning modules tailored to user needs. Lastly, regarding the delivery phase, we are still conducting iterative user testing to continue refining our prototype into an accessible, practical, and useful user-centric informative app as we prepare to bring the app to market.

Moving forward, our next steps are focused on polishing the final prototype based on tester feedback, conducting another round of more diverse user testing, and potentially marketing our product to organizations or agencies like the Department of Education that would recognize the value added to digital literacy efforts among younger demographics that have often have less cybersecurity awareness.

A complete record of our process, work products, and prototype development is available on our project website linked below.

Website: <https://adrienr4.wixsite.com/datagear>